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Title of Research Report

Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

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Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

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Abstract

We present SIMON, a hybrid integrator combining a second-order leapfrog scheme with a learned scalar pairwise force correction and two physical constraints: an analytic force direction that preserves Newtonian radial symmetry, and adaptive sub-stepping that resolves close gravitational encounters. Through a controlled four-run ablation study—removing each component in isolation—we show that long-horizon orbital stability over 100 years depends on these structural constraints: adaptive sub-stepping eliminates the timestep-frontier ejection failure (105× worst-case frontier error reduction), while the learned-direction baseline demonstrates, for the tested initial condition, that replacing analytic direction with a neural approximation produces unbounded trajectories. The neural network correction itself provides an 8.5% accuracy improvement without contributing to stability. Evaluated across seven distinct three-body initial conditions spanning near-circular to tight-binary dynamical regimes, SIMON maintains bounded dynamics for all stable configurations tested (mean divergence rate $\lambda = 0.092 \pm 0.068\ \mathrm{yr}^{-1}$), and agrees with the reference integrator on the occurrence of ejection in the three dynamically unstable configurations; post-ejection separations differ, as expected, because chaotic divergence amplifies integrator differences after the system becomes unbound. Energy drift scales as dt^2, consistent with second-order leapfrog theory, with negligible contribution from the neural correction. At the operational timestep ($dt = 0.04$ yr), SIMON achieves a 1.12× speedup over `ias15` with simulation performance invariant across five independent training seeds. These results demonstrate that long-horizon stability in hybrid neural integrators for chaotic gravitational systems requires structurally enforced physical constraints rather than learned approximations alone.

Keywords: Three-Body Problem; Neural Networks; Hybrid Integrator; Leapfrog Integration; Adaptive Sub-Stepping; Chaotic Dynamics; Gravitational N-Body Simulation; Lyapunov Divergence

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1 Introduction

The Three-Body Problem—the gravitational dynamics of three mutually interacting massive bodies—is a classical problem in physics because its trajectories are generally chaotic and highly sensitive to initial conditions. Although exact analytical solutions exist for two-body systems, long-horizon three-body prediction must usually be performed numerically. This remains important in astrophysical applications such as exoplanet stability, stellar cluster dynamics, and spacecraft trajectory design Gardner et al. [2006].

High-accuracy numerical integrators such as `ias15` provide reliable reference solutions, but their step-by-step structure makes long simulations computationally expensive Rein and Spiegel [2014]. Neural or hybrid methods offer a possible route to faster simulation Ulibarrena et al. [2024], but unconstrained learned force models can lose physical stability during close encounters, where small position or force errors are strongly amplified Dehnen [2001]. The central issue is therefore not simply whether a neural network can approximate a trajectory, but whether it can do so while preserving the physical structure needed for stable long-horizon dynamics.

This study addresses the following research question:

> *How closely can physically constrained neural network force models approximate the long-term stability of traditional numerical integrators in three-body simulations, while reducing computational cost without sacrificing qualitative dynamical stability?*

To answer this question, we present `SIMON`, a hybrid integrator that combines a second-order leapfrog scheme with a learned scalar pairwise force correction, analytically enforced Newtonian force direction, and adaptive sub-stepping during close encounters. The method is evaluated through controlled ablations that isolate which components are responsible for stability, accuracy, physical consistency, and computational performance.

The key contributions of this work are:

1. **A stability decomposition of hybrid neural integration:** controlled ablations show that adaptive sub-stepping and analytic force direction are the main stability mechanisms, while the neural correction primarily improves accuracy.
2. **A generalisation test across dynamical regimes:** `SIMON` is evaluated on seven distinct initial conditions, including stable, tight-binary, hierarchical, and physically unstable configurations.
3. **A physical-consistency and reproducibility analysis:** energy drift is shown to scale as dt^2, consistent with leapfrog theory, and results remain stable across five independent training seeds.

4. **A speed–accuracy frontier:** at the operational timestep, `SIMON` achieves a modest computational-cost reduction while preserving bounded dynamics across all tested stable configurations.

The paper is structured as follows. Section 2 reviews the physics background of three-body dynamics, chaos, and long-horizon prediction. Section 3 discusses computational modelling, learned simulators, and the research gap addressed by `SIMON`. Section 4 defines the `SIMON` architecture, training procedure, evaluation strategy, and scope. Section 5 presents experimental results across four themes: stability decomposition, generalisation, physical consistency, and computational performance. Section 6 discusses implications, limitations, and future directions.

2 Literature Review

The literature relevant to this project lies at the intersection of celestial mechanics, numerical integration, and scientific machine learning. Classical mechanics explains why the unrestricted three-body problem is sensitive to initial conditions and difficult to solve analytically, while numerical-analysis literature shows that long-horizon behaviour depends strongly on whether an algorithm preserves geometric structure and handles close encounters robustly Hairer et al. [2006], Rein and Tamayo [2015], Rein and Spiegel [2014], Boekholt and Portegies Zwart [2015]. More recent machine-learning work shows that neural networks can act as learned simulators or learned Hamiltonian surrogates, but also that their success depends on the physical inductive biases built into the architecture Chen et al. [2018], Battaglia et al. [2016], Greydanus et al. [2019], Gruver et al. [2022], Horn et al. [2025]. This section gives the physics background needed to motivate the computational gap addressed by `SIMON`.

2.1 From Two-Body Solvability to Three-Body Non-Integrability

The Two-Body Problem is analytically tractable because the mutual gravitational attraction between two isolated point masses can be reduced to an equivalent one-body problem in a central potential. Newton's law of universal gravitation gives the magnitude of the force as

$$F = G\frac{m_1 m_2}{r^2}, \tag{1}$$

where G is the gravitational constant, m_1 and m_2 are the two masses, and r is their separation. This framework explains the classical Keplerian orbits used to model systems such as planets, moons, and satellites Murray and Dermott [1999].

The unrestricted Three-Body Problem is fundamentally different. Each body is accelerated by the other two at the same time, so the equations of motion are coupled and nonlinear. For bodies with positions $\mathbf{r}_i$ and masses m_i, the Newtonian equation of motion can be written compactly as

$$\ddot{\mathbf{r}}_i = G \sum_{j \neq i} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}, \qquad i = 1, 2, 3. \tag{2}$$

Unlike the two-body case, this system has no general closed-form solution. Its behaviour depends sensitively on mass ratios, separations, and velocities, making numerical simulation the standard approach for studying generic three-body dynamics Murray and Dermott [1999].

2.2 Chaos and Long-Horizon Prediction

A central difficulty of the Three-Body Problem is chaos. Small differences in initial position or velocity can grow rapidly, causing two initially similar trajectories to diverge over long times. One common way to describe this behaviour is the Lyapunov exponent λ, where an initial separation $\delta(0)$ grows approximately as

$$\delta(t) \approx \delta(0) e^{\lambda t}. \tag{3}$$

A positive value of λ indicates exponential sensitivity to initial conditions Ott [2002]. This idea traces back to Poincaré's analysis of the three-body problem, which showed that gravitational systems can exhibit intricate, non-repeating trajectories rather than simple closed-form motion Poincaré [1890].

For this project, chaos has an important methodological consequence. In a chaotic system, long-term pointwise trajectory agreement is not always the right standard for judging a simulator, because even small numerical differences can produce phase drift. Instead, a useful evaluation must also consider qualitative dynamical behaviour: whether a system remains bounded, whether ejection occurs, whether energy drift is physically explainable, and whether divergence grows at a controlled rate. This motivates the use of divergence-rate proxies and boundedness checks later in this paper.

2.3 Restricted and Unrestricted Three-Body Problems

The Restricted Three-Body Problem is a simplified version in which two massive bodies influence a third body whose mass is small enough to neglect. This formulation led to important results such as Euler's and Lagrange's special solutions and the five Lagrange points Bistafa [2021], Lagrange [1772]. These ideas are important in practical mission design; for example, the James Webb Space Telescope operates near the Sun–Earth L_2 region Gardner et al. [2006].

However, the present work focuses on the unrestricted problem, where all three bodies exert mutual gravitational forces. This case is more directly relevant to testing neural-assisted integrators because the system does not reduce to a fixed background potential and can include strong pairwise encounters. The unrestricted setting therefore provides a stricter test of whether a learned force model can remain dynamically usable when all interactions are active.

The key implication for this project is that unrestricted three-body dynamics combine three features that make long-horizon simulation difficult: nonlinear mutual forcing, sensitivity to initial conditions, and close encounters where the Newtonian force changes rapidly with separation. These features make purely pointwise long-term trajectory matching unreliable and motivate the computational question addressed next: how can a hybrid neural integrator reduce computational cost while preserving the physical structure needed for stable qualitative dynamics?

3 Computational Modelling

Since unrestricted three-body dynamics do not have a general closed-form solution, their evolution must be computed numerically. A traditional integrator advances the system step by step, so long simulations require many force evaluations, and close encounters can amplify small numerical errors into large trajectory differences. High-accuracy tools such as `REBOUND` and `ias15` are therefore essential benchmarks for this project, because they show what reliable long-horizon gravitational integration should look like Rein and Liu [2012], Rein and Spiegel [2014]. At the same time, numerical-analysis literature shows that long-time reliability is not determined only by local truncation error: preserving geometric structure and resolving close interactions are central to physical reliability Hairer et al. [2006], Boekholt and Portegies Zwart [2015].

Neural networks offer a possible route to reducing this computational burden, but the role of the network must be defined carefully. Some learned-surrogate approaches train a model to map from an initial state directly to a future state, thereby skipping some or all intermediate integration steps. `SIMON` does not use this direct state-to-future-state strategy. It remains a timestep-based hybrid integrator, using a neural network only as a local scalar force-correction mechanism during close encounters. The aim is therefore not to replace numerical integration entirely, but to test whether a learned component can reduce computational cost while preserving the physical structure needed for stable long-horizon rollouts.

This distinction is important because close encounters are the most fragile part of the problem. Under Newtonian gravity, the force between two bodies separated by distance r is proportional to $1/r^2$, so the force magnitude and its gradient change rapidly when r

becomes small. A small learned error in position, force magnitude, or force direction can then produce a large artificial impulse, causing non-physical energy change or ejection-like behaviour Dehnen [2001]. The central modelling question is therefore not simply whether a neural network can approximate a three-body trajectory, but which parts of the Newtonian force law must remain exact for a learned integrator to remain dynamically usable.

3.1 Learned Simulators and Research Gap

Prior work gives three important lessons for this project. First, classical integrators provide the reliability standard: `WHFast` demonstrates the value of fast symplectic long-term orbit integration, while `ias15` provides a high-order adaptive Gauß–Radau reference method for gravitational dynamics Rein and Tamayo [2015], Rein and Spiegel [2014]. Work on the reliability of N-body simulations further shows that chaotic three-body scattering can require very high numerical precision to obtain converged individual trajectories Boekholt and Portegies Zwart [2015]. This makes close encounters and long-horizon agreement especially demanding tests for any hybrid method.

Second, learned simulators show that neural networks can approximate dynamical systems, but they do not by themselves guarantee physical reliability. Neural ODEs learn a continuous-time derivative field Chen et al. [2018], while Interaction Networks and graph-network simulators learn object interactions and roll out multi-body systems over time Battaglia et al. [2016], Sanchez-Gonzalez et al. [2020]. In celestial mechanics, Breen *et al.* showed that deep networks trained on high-precision three-body solutions can approximate bounded-time three-body evolution at fixed cost Breen et al. [2020]. Physics-informed neural networks take another route by imposing governing equations through the loss function Cuomo et al. [2022], including recent applications to the three-body problem Pereira et al. [2025]. These works show the promise of learned dynamics, but they do not isolate which components of the Newtonian force law should be learned and which should remain analytically enforced.

Third, structure-preserving machine learning shows that adding physical inductive bias can improve extrapolation. Hamiltonian and Lagrangian Neural Networks encode mechanical structure through energy-based formulations Greydanus et al. [2019], Cranmer et al. [2020], while symplectic or Hamiltonian-inspired architectures aim to preserve geometric properties of the underlying dynamics Zhong et al. [2020], Jin et al. [2020], Tong et al. [2021], David and Méhats [2023], Horn et al. [2025]. However, later analysis also shows that not every physics-inspired bias contributes equally; some apparent Hamiltonian-network advantages may arise from other inductive biases such as second-order structure Gruver et al. [2022]. This supports a more targeted question: in a chaotic gravitational system, which physical constraints are actually necessary for stable long-horizon rollouts?

This question is especially important because exact pointwise agreement over long horizons is often unrealistic in chaotic systems. A useful learned model should remain in the correct qualitative dynamical regime, rather than only minimising short-horizon rollout error Linot et al. [2022], Margazoglou and Magri [2023]. For the three-body problem, this means preserving bounded motion when the reference solution remains bounded, identifying ejection when the reference system becomes unstable, and keeping energy drift physically interpretable.

The gap addressed by `SIMON` is therefore specific: prior work shows that learned simulators can be fast and that physical inductive biases can improve generalisation, but it does not clearly test whether a neural force model should learn the full gravitational force vector or whether Newtonian radial direction should remain analytic while learning is restricted to a lower-dimensional scalar correction. Nor does it usually separate the stabilising effect of close-encounter timestep refinement from the effect of the neural model itself. `SIMON` is designed around this gap. It combines a leapfrog backbone with analytically enforced radial force direction, adaptive sub-stepping during close encounters, and a neural network limited to a scalar pairwise force correction. Its novelty is methodological: through controlled ablation, it tests which physical constraints are necessary for dynamically usable long-horizon three-body rollouts.

4 SIMON: Architecture and Methodology

4.1 Method Overview and Reference Solver

Building on the research gap identified in Section 3, this section defines the hybrid architecture used to test whether a neural force correction can be made dynamically usable in the unrestricted Three-Body Problem. `SIMON` is a timestep-based integrator: it does not predict the final state directly from the initial state. Instead, it evolves the system step by step using a leapfrog backbone, while using a neural network only as a local scalar correction to the pairwise force magnitude during close encounters.

The reference solution throughout this paper is generated using `REBOUND` [Rein and Liu, 2012], specifically the `ias15` integrator [Rein and Spiegel, 2014]. `ias15` is a high-order adaptive Gauß–Radau method designed for gravitational dynamics and is used here as the benchmark for long-horizon trajectory comparison, boundedness/ejection classification, and runtime comparison. `SIMON` is not intended to replace `ias15` as a universal reference solver. Its purpose is to test a narrower methodological question: which physical constraints must be enforced for a neural-assisted integrator to preserve qualitative long-horizon dynamics?

The method has three core design choices:

1. a second-order leapfrog update provides the integration backbone;
2. Newtonian radial force direction is computed analytically at every step;
3. the neural network learns only a scalar correction to the softened pairwise force magnitude during close encounters.

This separation between learned and analytic components allows the ablation study in Section 5.1 to identify which parts of the method are responsible for stability, accuracy, and computational performance.

4.2 Scalar Force-Correction Model

For bodies with positions $\mathbf{x}_i$, velocities $\mathbf{v}_i$, and masses m_i, pairwise interactions are computed using the displacement

$$\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i, \qquad r_{ij} = \|\mathbf{r}_{ij}\|.$$

The exact Newtonian pairwise force is

$$\mathbf{F}_{ij}^{\text{Newt}} = G\frac{m_i m_j}{r_{ij}^3}\mathbf{r}_{ij},$$

where the vector direction is determined entirely by the geometry of the two bodies Murray and Dermott [1999]. This exact radial direction is preserved in `SIMON`; the neural network is never allowed to learn or rotate the force direction.

Close encounters are numerically difficult because the Newtonian force magnitude changes rapidly as r_{ij} becomes small. To obtain a smooth local correction target, `SIMON` first computes a softened pairwise force using

$$\tilde{r}_{ij} = \sqrt{r_{ij}^2 + \epsilon^2}, \qquad \mathbf{F}_{ij}^{\text{soft}} = G\frac{m_i m_j}{\tilde{r}_{ij}^3}\mathbf{r}_{ij}.$$

The network then predicts a scalar correction factor c_{ij} so that

$$\mathbf{F}_{ij}^{\text{SIMON}} = c_{ij}\mathbf{F}_{ij}^{\text{soft}}.$$

Equivalently, the ideal correction satisfies

$$c_{ij} = \left(\frac{\tilde{r}_{ij}}{r_{ij}}\right)^3, \qquad \log c_{ij} = 3\log\left(\frac{\tilde{r}_{ij}}{r_{ij}}\right).$$

The model is therefore trained to predict $\log c_{ij}$ from the feature vector

$$z_{ij} = [\log \tilde{r}_{ij}, \log m_i, \log m_j].$$

This keeps the learned task one-dimensional in the output while preserving the exact Newtonian force direction. The architecture is a small multilayer perceptron,

$$3 \to 32 \to 32 \to 32 \to 1,$$

with SiLU activations. This compact design is sufficient because the learned function is a smooth scalar correction, not a full vector-valued force law.

4.3 Hybrid Integration and Adaptive Sub-Stepping

At each macro timestep, `SIMON` computes all pairwise forces and updates the state using a velocity-Verlet/leapfrog step. The force calculation is divided into three regimes:

1. **Far field:** if $r_{ij} > 500\epsilon = 0.15\,\mathrm{AU}$, the NN is not invoked and the pairwise force is computed using the exact Newtonian expression.
2. **Close encounter:** if $r_{ij} < 0.15\,\mathrm{AU}$, the NN predicts the scalar correction c_{ij} applied to the softened force.
3. **Safety fallback:** if the encounter is extremely close ($\tilde{r}_{ij} < 5 \times 10^{-4}\,\mathrm{AU}$), or if the predicted correction lies outside the gate $0.2 \leq c_{ij} \leq 5.0$, the NN output is rejected and the exact Newtonian force is used instead.

This design follows the principle of hybrid scientific machine learning: learned components are used only where they are intended to help, while physically unsafe outputs are replaced by analytic force evaluations Ulibarrena et al. [2024]. In particular, the NN correction changes only the scalar magnitude of the force; the direction remains the exact radial Newtonian direction throughout.

To resolve close encounters without globally shrinking the timestep, `SIMON` uses adaptive sub-stepping. At each macro step, the minimum pairwise distance $r_{\min}$ is computed. If $r_{\min} < 0.05\,\mathrm{AU}$, the macro step dt is divided into

$$n_{\mathrm{sub}} = \min\left(16,\ \max\left(2, \left\lceil \frac{0.05}{r_{\min}} \right\rceil\right)\right)$$

sub-steps, each of length dt/n_{sub}. Thus, closer encounters receive finer temporal resolution, while the maximum of 16 sub-steps limits computational overhead. All positions, velocities, accelerations, and forces are stored and updated in float64; the neural-network forward pass uses float32 weights, with the scalar correction converted back to float64 before being applied.

4.4 Training Procedure

Training data is generated from synthetic pairwise configurations sampled across the separation range relevant to the force correction. The target is the analytic correction $\log c_{ij} = 3\log(\tilde{r}_{ij}/r_{ij})$, so the network learns how to convert the softened force magnitude back toward the Newtonian force magnitude while keeping the force direction analytic. Approximately 60% of samples are drawn from the transition region $r \in [0.5\epsilon, 50\epsilon]$, where softening changes the force appreciably, and 40% from the far field $r \in [50\epsilon, 10]$, where the ideal correction is close to one.

Table 1: Training configuration for the scalar force-correction network.

Component	Setting
Input features	$[\log \tilde{r}_{ij}, \log m_i, \log m_j]$
Target	$\log c_{ij} = 3\log(\tilde{r}_{ij}/r_{ij})$
Architecture	$3 \to 32 \to 32 \to 32 \to 1$ with SiLU activations
Training samples	40,000
Validation samples	10,000
Optimizer	AdamW, weight decay 1×10^{-5}
Learning rate schedule	10^{-3} with cosine annealing to 10^{-5} Loshchilov and Hutter [2017]
Loss	Mean squared error on $\log c_{ij}$
Epochs	5,000
Primary training seed	42
Training device	NVIDIA GeForce RTX 5050 CUDA 13.0
Best validation MSE	1.1×10^{-5}

Seed robustness is tested separately in Section 5.3.3 by retraining the same architecture with five independent random initialisations.

4.5 Evaluation Strategy and Scope

Because three-body trajectories are chaotic, long-horizon pointwise agreement alone is not a sufficient evaluation standard. `SIMON` is therefore evaluated using both quantitative and qualitative diagnostics:

Table 2: Evaluation diagnostics used to assess SIMON.

Diagnostic	Purpose
Trajectory overlays	Visual comparison of SIMON and ias15 trajectories, including phase drift.
RMS separation	Measures accumulated position deviation from the ias15 reference.
Divergence-rate proxy λ	Summarises early-to-mid horizon separation growth in chaotic rollouts.
Bounded/ejected status	Tests whether SIMON preserves the qualitative dynamical regime.
Energy drift	Checks whether physical consistency follows the expected leapfrog dt^2 scaling.
Seed robustness	Tests whether results depend on random NN initialisation.
Speed–accuracy frontier	Compares runtime gain against trajectory error across timesteps.

The fitted value of λ used in this work is a divergence-rate proxy rather than a formal Lyapunov exponent [Bishop, 2025]. It reflects both the intrinsic chaotic sensitivity of the three-body system and the accumulated model/integration mismatch between SIMON and ias15. A perfect reproduction of the reference trajectory would give $\lambda = 0$ in this comparison, while larger values indicate faster separation from the reference rollout.

The scope of the method is deliberately limited. SIMON is tested as a controlled methodological study for unrestricted three-body systems, not as a universal replacement for ias15. Its purpose is to determine whether a neural force correction can remain useful when exact radial direction and close-encounter timestep refinement are enforced analytically. The limitations of this scope are discussed further in Section 6.3.

5 Results

In this section we present the experimental results for SIMON across four themes. Section 5.1 isolates the individual contribution of each architectural component through a controlled four-run ablation. Section 5.2 evaluates generalisation across seven initial conditions. Section 5.3 examines physical consistency and reproducibility. Section 5.4 characterises the speed-accuracy trade-off as a separate computational-performance result.

Throughout this section, λ denotes the fitted divergence-rate proxy (slope of $\log \delta(t)$ in the 10–50 yr window), and "bounded" indicates that all three bodies remain within 10 AU of the system barycentre at $T = 100$ yr. The reference integrator is ias15 [Rein and Spiegel, 2014]; all initial conditions use masses [1.0, 0.01, 0.005] $M_\odot$ unless otherwise stated. At the operational timestep $dt = 0.04\,\text{yr}$, $\lambda = 0.1655\,\text{yr}^{-1}$ and final RMS = 1.54 AU for the default initial condition (IC1).

5.1 Results I — Stability Decomposition

5.1.1 The Four-Run Ablation Design

To isolate the role of each architectural component, we conduct a controlled four-run ablation in which exactly one component is removed or altered per run while the initial condition IC1, timestep grid, simulation horizon, and evaluation protocol remain fixed. Runs A, C, and D use the same trained SIMON weights; Run B uses a separately trained Vector NN baseline because the architecture itself is changed to learn force direction. The four runs are summarised in Table 3.

Table 3: Four-run ablation framework for IC1 over $T = 100\,\text{yr}$. Each run removes or replaces one component of the full `SIMON` configuration.

Run	Adaptive stepping	Force direction	NN correction	Outcome
A	**OFF**	Analytic	ON	Ejection (161.7 AU)
B	ON	**Learned (Vector NN)**	N/A	Ejection (83.0 AU)
C	ON	Analytic	**OFF**	Bounded (1.67 AU)
D (SIMON)	ON	Analytic	ON	Bounded (1.54 AU)

The logic of the design is as follows. Runs A and B establish stability requirements by showing that removing a component causes ejection. Run C establishes the NN's role by showing that its removal leaves the system stable. Run D is the full system, providing the accuracy ceiling against which Run C is compared. Because Run B is evaluated on the default initial condition, its trajectory outcome is interpreted together with the gate-sensitivity analysis rather than as a standalone statistical generalisation.

5.1.2 Run A: Adaptive Sub-Stepping Is the Primary Stability Mechanism

Without adaptive sub-stepping, `SIMON` uses a fixed leapfrog step for every integration step regardless of body separation. At $dt = 0.01\,\text{yr}$, a close encounter occurs whose duration is shorter than the fixed timestep can resolve. The lightest body receives an incorrectly large gravitational impulse in a single step and is ejected to 161.7 AU at $T = 100\,\text{yr}$. With adaptive sub-stepping active, the same close encounter is resolved by splitting the step into 2 to 16 sub-steps when $r_{\min} < 0.05$ AU, and ejection is eliminated across all tested timesteps for IC1.

Figure 1 shows the complete frontier sweep across five timestep values: $dt = 0.005$, 0.01, 0.02, 0.04, and 0.08 yr. The sweep is run for both conditions. The worst-case final RMS across the frontier reduces from 161.7 AU before adaptive sub-stepping ($dt = 0.01$ yr) to 1.54 AU after adaptive sub-stepping ($dt = 0.04$ yr), a 105× improvement.

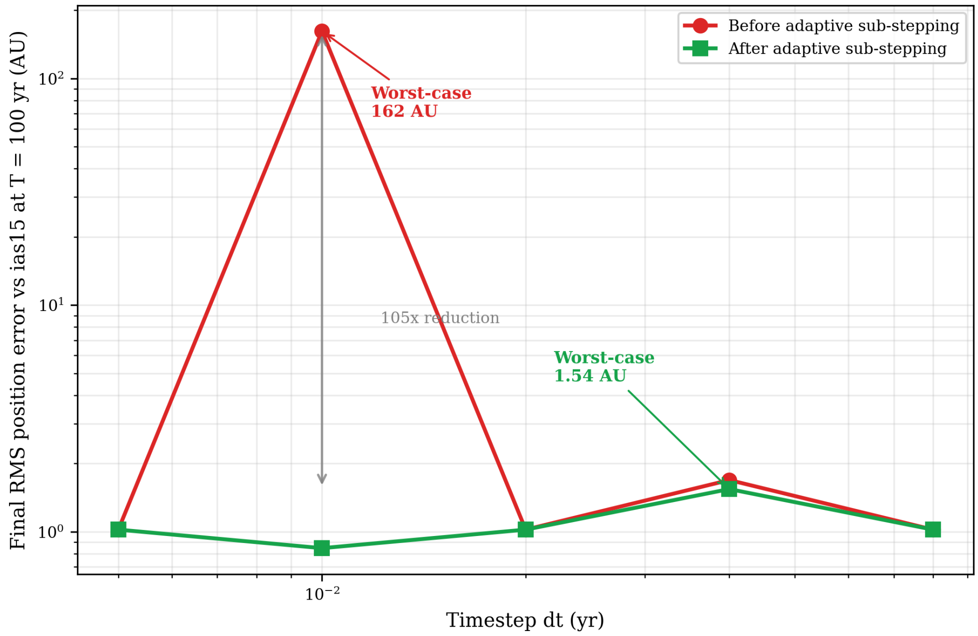

Figure 1: Final RMS position error vs. timestep for `SIMON` with and without adaptive sub-stepping on IC1.

At the operational timestep $dt = 0.04$ yr—the main timestep used for the operational comparison—adaptive sub-stepping costs +28% per-step overhead (19.21 → 24.57 μs), reducing speedup from 1.53× to 1.12× relative to `ias15`, while reducing final RMS from 1.691 to 1.543 AU (8.8% improvement) and the divergence-rate proxy from 0.1712 to 0.1655 yr^{-1} (3.3% improvement). Both configurations remain faster than `ias15`. The reported NN invocation rate is less than 1.1% of pair evaluations involving close encounters with $r < 0.15$ AU. Adaptive sub-stepping uses the stricter condition $r_{\min} < 0.05$ AU, so it is triggered only on a subset of those close-encounter cases, leaving the majority of steps at the base leapfrog cost.

5.1.3 Run B: Analytic Force Direction as a Structural Physical Constraint

In Run B, the analytic force direction $\mathbf{u}_{ij} = \mathbf{r}_{ij}/\|\mathbf{r}_{ij}\|$ is replaced by a learned direction from a Vector NN. The Vector NN uses the same hidden-layer structure as `SIMON` but receives strictly more information, including the geometric direction components $[r_x/r, r_y/r, r_z/r]$, and predicts a unit force-direction vector. Post-training verification confirms sub-degree directional accuracy, with maximum directional error approximately 0.04° across the tested distance range. For the tested initial condition, however, the learned-direction architecture produces unbounded trajectories, reaching final RMS error 83.0 AU at $T = 100$ yr, while `SIMON` remains bounded at 1.54 AU. In plain terms, this run tests whether gravity's direction should be learned approximately or enforced exactly from geometry.

To test whether this unbounded outcome was caused by a single anomalous direction prediction or by accumulated small errors, the Vector NN direction gate was tightened and the simulation was re-run.

Table 4: Gate-sensitivity test for the learned-direction Vector NN baseline. Tightening the direction gate does not prevent the unstable outcome.

Gate setting	Interpretation	Outcome
$\cos(\theta) < 0.866$	Standard 30° gate; triggers only on large direction deviations.	Ejection unchanged; final RMS 83.0 AU.
$\cos(\theta) < 0.9999985$	Much tighter than the standard gate; close to the Vector NN's measured sub-degree accuracy.	Identical ejection timing; final RMS 83.0 AU.

The gate-sensitivity result shows that the mechanism is not a single catastrophic misprediction, but the accumulation of small non-radial impulses over many steps. When the gate is tightened to $\cos(\theta) < 0.9999985$, corresponding to an angular threshold of approximately 0.1°, the ejection timing and final RMS remain unchanged. This threshold is far tighter than the original 30° gate and close to the Vector NN's measured sub-degree directional accuracy.

Thus, the failure is not an outlier that a gate can catch. Instead, many individually acceptable direction errors accumulate during close encounters. A simple scale estimate illustrates the issue: at $dt = 0.04\,\text{yr}$ over 100 yr, there are approximately 2,500 integration steps, and 0.04° per close-encounter step can accumulate to roughly 100° of non-radial impulse. Because Newtonian gravity is a central force directed exactly along the line connecting two bodies, any learned perpendicular component has no physical basis and can compound into non-physical energy injection in a chaotic system.

Figures 2 and 3 illustrate the physical consequence of the gate-sensitivity finding. During the bounded phase, `SIMON` and the Vector NN have nearly identical fitted divergence rates (0.1655 vs. $0.1656\,\text{yr}^{-1}$), but the learned-direction model later becomes unbounded for this tested initial condition, reaching final RMS error 83.0 AU. In Figure 3, the long red dotted segments are escape paths of the Vector NN trajectory, not stable orbital loops.

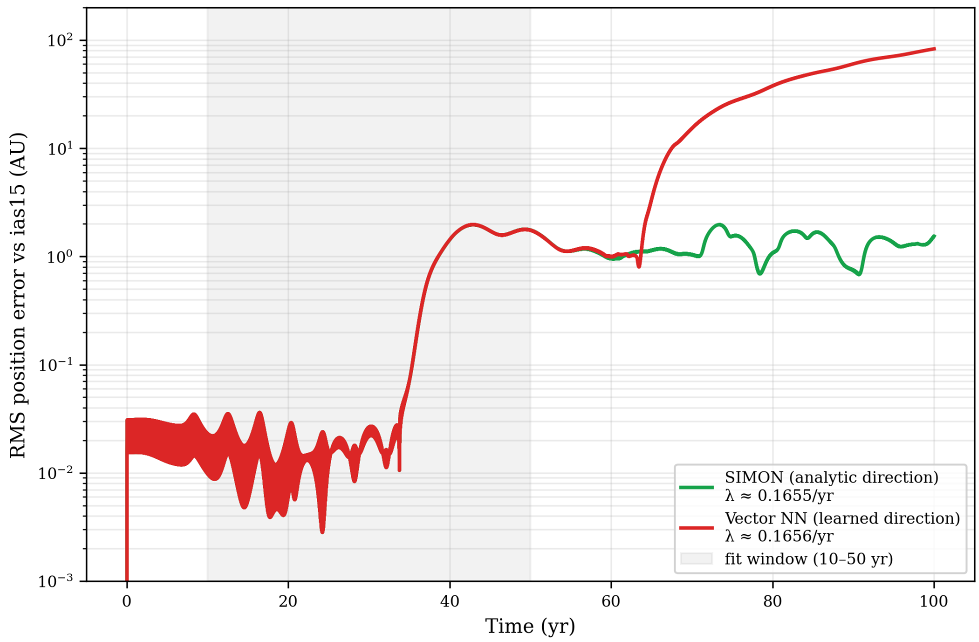

Figure 2: RMS position error vs. time for `SIMON` and the Vector NN baseline. The shaded region marks the 10–50 yr divergence-rate fit window.

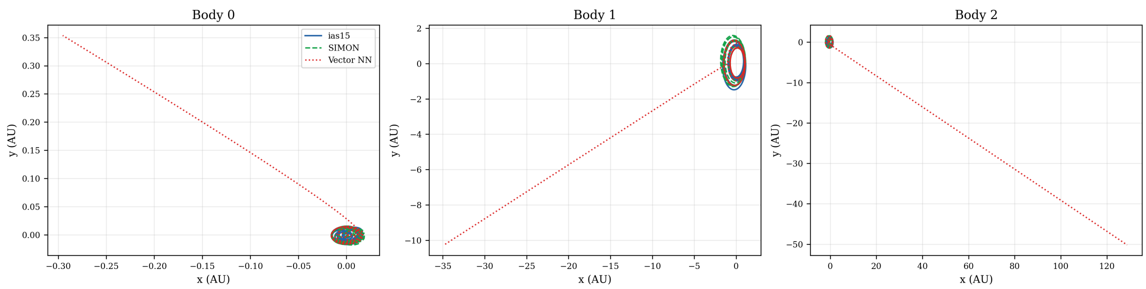

Figure 3: XY trajectory overlays for `ias15`, `SIMON`, and the Vector NN over 100 years. Each body is shown separately with independent axis limits.

This result supports the conclusion that analytic force direction is not merely a convenience but a structural stability constraint. For the tested initial condition, the learned-direction baseline becomes unbounded despite sub-degree angular accuracy, and the gate-sensitivity analysis shows that the outcome is insensitive to the specific gate threshold chosen. Analytic direction therefore removes, by construction, a non-physical perpendicular-force error mode that a learned direction model can still introduce during close encounters.

5.1.4 Run C: The Neural Network Correction Is an Accuracy Mechanism

In Run C, the NN correction is disabled entirely: the correction factor c is fixed at 1.0 for all pair evaluations, reducing the force to pure softened Newtonian gravity. The analytic force direction and adaptive sub-stepping remain active. The result is a bounded simulation at 1.67 AU—not an ejection.

Figure 4 shows that Run C (No-NN baseline, red dashed) and Run D (full `SIMON`, green solid) are visually indistinguishable throughout the 100-year horizon. Both achieve $\lambda = 0.1655\,\mathrm{yr}^{-1}$ to four decimal places. The NN correction contributes 8.5% improvement in final RMS accuracy ($1.67 \rightarrow 1.54$ AU) at the 0.45% of pair evaluations involving close encounters below 0.15 AU, where it is invoked.

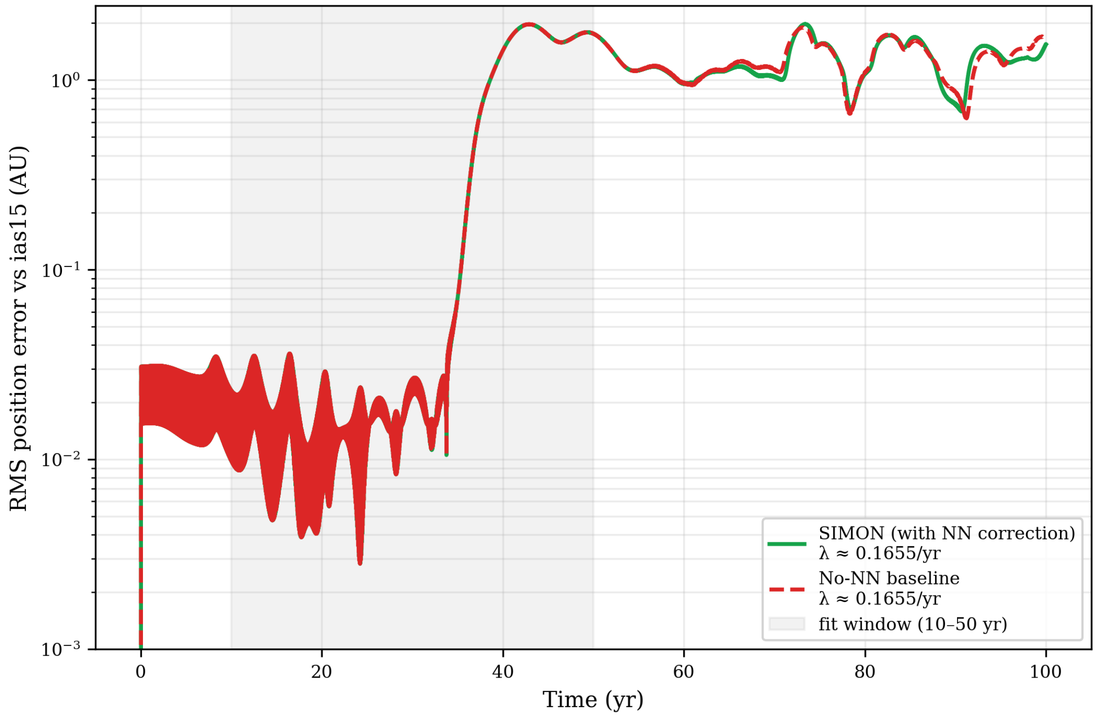

Figure 4: RMS position error vs. time for full `SIMON` and the No-NN baseline. Both runs retain analytic direction and adaptive sub-stepping.

This result clarifies the role of the NN correction. Run A shows that close-encounter stability is provided by adaptive sub-stepping, since removing that mechanism causes ejection even when the NN correction remains active. Run C shows that removing the NN correction while retaining adaptive sub-stepping leaves the system bounded, but increases final RMS error from 1.54 AU to 1.67 AU. Therefore, the NN correction is best interpreted as a targeted accuracy refinement at close-encounter steps, rather than the primary stability mechanism.

5.1.5 Synthesis: The Two-Constraint Principle

The four-run ablation establishes a clear decomposition of `SIMON`'s performance:

- **Adaptive sub-stepping** eliminates the leapfrog ejection failure at $dt = 0.01$ yr for IC1 (Run A: 161.7 AU without sub-stepping versus 0.847 AU with sub-stepping).
- **Analytic force direction** is a structural physical constraint (Run B: for the tested

initial condition, replacing analytic direction with a learned direction produces unbounded trajectories with final RMS 83.0 AU).

- **NN correction** is an accuracy mechanism (Run C: removing it leaves stability intact but costs 8.5% accuracy).
- **Full** `SIMON` (Run D) achieves the best accuracy by combining all three components.

Figure 5 shows the XY trajectory overlay and coordinate time series for the full `SIMON` system (Run D), confirming bounded dynamics for all three bodies over 100 years.

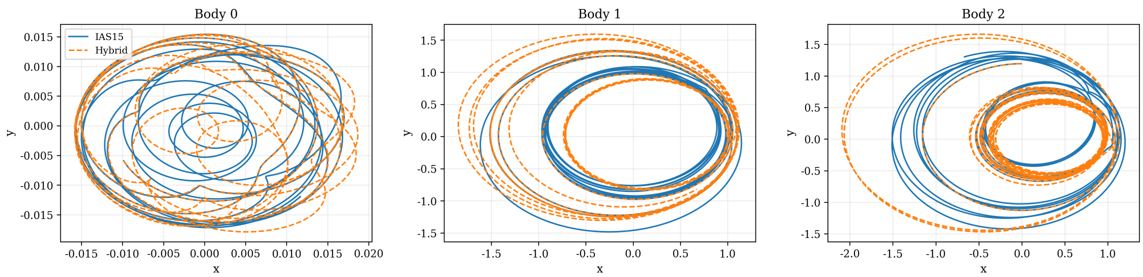

Figure 5: XY trajectory overlays for all three bodies comparing `SIMON` with `ias15` over 100 years.

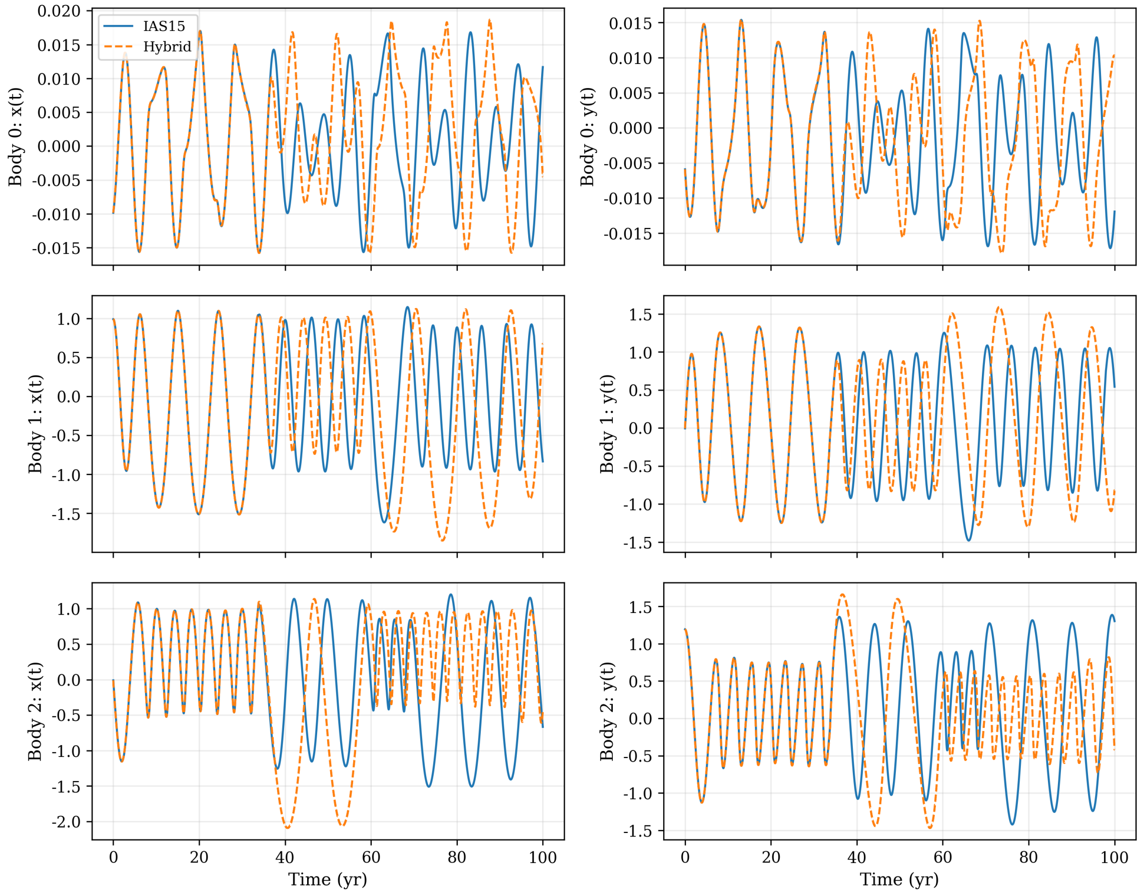

Figure 6: Coordinate time series $x(t)$ and $y(t)$ for all three bodies over 100 years. Phase drift accumulates while the trajectories remain bounded.

The primary error mode is phase drift—a gradual timing mismatch that accumulates while preserving the qualitative orbital geometry and boundedness of all three bodies. This is consistent with the chaotic sensitivity of the Three-Body Problem [Ott, 2002] and does not indicate structural failure of the integrator.

5.2 Results II — Generalisation Across Initial Conditions

5.2.1 Seven Initial Conditions Spanning the Dynamical Spectrum

To test whether `SIMON`'s performance extends beyond the single default configuration (IC1) used in the ablation study, we evaluate across seven distinct three-body initial conditions designed to span qualitatively different dynamical regimes. Table 5 summarises the configuration and simulation outcomes for all seven ICs. All runs use masses $[1.0, 0.01, 0.005]\,M_\odot$ except IC2 (near-equal mass), $T = 100\,\text{yr}$, and $dt = 0.04\,\text{yr}$.

Table 5: Multi-IC evaluation results at $T = 100\,\text{yr}$ and $dt = 0.04\,\text{yr}$. Summary statistics are computed over the four bounded configurations only.

IC	Regime	$\boldsymbol{\lambda}$ (yr^{-1})	RMS (AU)	Bounded	Speedup
IC1: Default	Moderate scattering	0.1655	1.54	Yes	1.12×
IC2: Near-equal	Strongly interacting	0.0430	116.9	No	1.77×
IC3: Tight pair	High encounter rate	0.1207	3.45	Yes	1.24×
IC4: Hierarchical	Weakly coupled outer	0.0990	1.60	Yes	1.14×
IC5: High ecc. (v1)	High eccentricity	0.0648	20.0	No	1.22×
IC6: Near-circular	Low chaos	−0.019	0.008	Yes	1.08×
IC7: High ecc. (v2)	High eccentricity	0.0341	11.4	No	1.11×
Mean (bounded ICs only)		0.092	1.65	–	–
Std (bounded ICs only)		0.068	1.22	–	–
Bounded count		–	–	4/7	–

5.2.2 Bounded Configurations: Four ICs

All four stable configurations maintain bounded dynamics for the full 100-year horizon. The configurations span the complete stable dynamical spectrum of the three-body problem:

- **IC6 (near-circular)**: $\lambda \approx 0$, final RMS $= 0.008\,\text{AU}$. The NN is never invoked (NN_frac $= 0.000$); `SIMON` computes exact Newtonian forces for every step. The residual 0.008 AU is the pure leapfrog–Gauss-Radau integrator difference.
- **IC4 (hierarchical)**: $\lambda = 0.099\,\text{yr}^{-1}$, final RMS $= 1.60\,\text{AU}$. The outer body at 5 AU is weakly coupled; the NN is also never invoked.
- **IC1 (default)**: $\lambda = 0.1655\,\text{yr}^{-1}$, final RMS $= 1.54\,\text{AU}$. The default IC1 baseline.

- **IC3 (tight binary)**: $\lambda = 0.121\,\text{yr}^{-1}$, final RMS $= 3.45\,$AU. The inner body at $0.5\,$AU generates continuous close encounters; despite this, `SIMON` remains bounded throughout.

Figure 7 summarises the divergence rates, final RMS values, and speedups across all seven ICs.

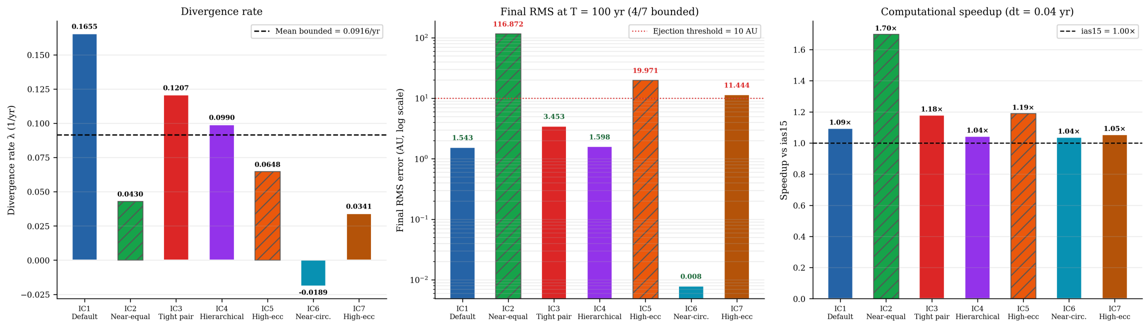

Figure 7: Summary across seven initial conditions: divergence rate, final RMS error, and speedup relative to `ias15`. Hatched bars denote physically unstable configurations.

Figure 8 shows the divergence curves for all seven ICs simultaneously.

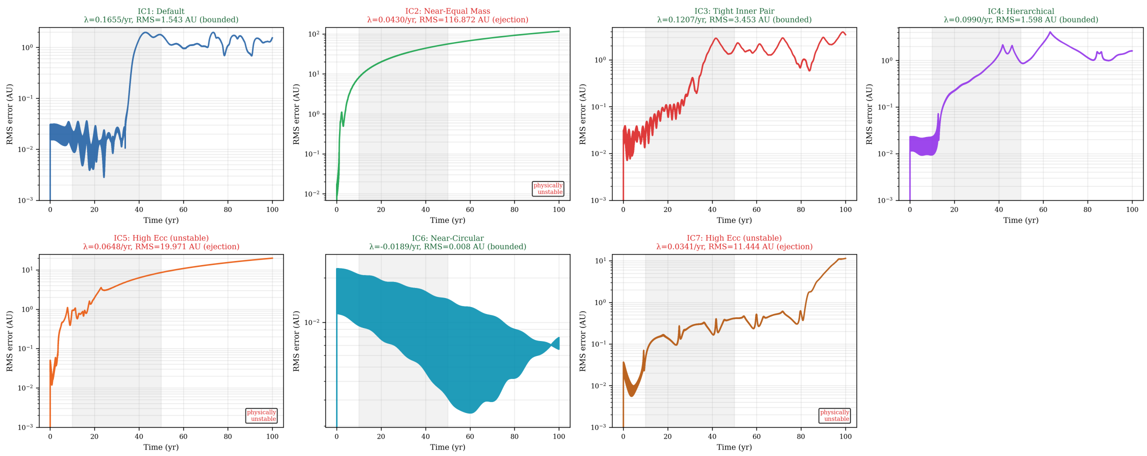

Figure 8: RMS position error vs. time for all seven initial conditions. The shaded region marks the 10–50 yr divergence-rate fit window.

5.2.3 Physically Unstable Configurations

Three configurations produce ejections (IC2, IC5, IC7). In each case, the reference integrator `ias15` independently also ejects a body, confirming these are physical instabilities rather than `SIMON` failures:

- **IC2 (near-equal mass)**: `SIMON` ejects to $116.9\,$AU; `ias15` ejects to $58.9\,$AU. The $2\times$ post-ejection discrepancy is consistent with chaotic amplification of integrator differences after the system becomes dynamically unbound.

- **IC5 (high eccentricity, $v = 2.2$, $r = 0.3$ AU)**: Both integrators eject; `ias15` ejects Body 2 at $t = 34.8$ yr.
- **IC7 (high eccentricity redesign, $v = 1.6$, $r = 0.6$ AU)**: Both integrators eject; `ias15` ejects Body 2 at $t = 88.4$ yr.

`SIMON` does not suppress these physical instabilities. This is expected and physically consistent: the model agrees with `ias15` on the occurrence of dynamical instability rather than artificially stabilising configurations that are physically unbound. The post-ejection distances are not expected to match exactly because chaotic divergence amplifies integrator differences after unbinding. High-eccentricity configurations with these mass ratios proved intrinsically unstable in both cases tested: the eccentric inner body transfers orbital energy to the outer body through repeated periapsis passages, eventually unbinding it regardless of integrator.

5.3 Results III — Physical Consistency and Reproducibility

5.3.1 Energy Conservation: Leapfrog Attribution

We track fractional energy drift $\Delta E(t)/E_0 = (E(t) - E_0)/|E_0|$ throughout each simulation, where $E(t) = \mathrm{KE}(t) + \mathrm{PE}(t)$ is the total mechanical energy. `ias15` conserves energy to float64 machine precision ($\approx 10^{-14}$%). `SIMON` at the operational timestep ($dt = 0.04$ yr) shows:

- IC1: 1.39% maximum fractional drift (NN_frac = 0.0045)
- IC3: 0.67% maximum fractional drift (NN_frac = 0.000)
- IC4: 13.77% maximum fractional drift (NN_frac = 0.000)

IC3 and IC4 both have NN_frac = 0.000: the NN correction was never applied during their simulations. Yet both show comparable energy drift to IC1. The NN correction cannot cause energy drift it never contributed to; the drift must arise from the leapfrog integrator. This is the attribution proof.

IC4's higher fractional drift (13.77%) reflects its small binding energy $|E_0| = 0.006$ (AU^2/yr^2)—the smallest of the three ICs tested. Because fractional drift is normalised by $|E_0|$, weakly bound hierarchical configurations can show larger relative energy drift at the same timestep. This should not be interpreted as NN-driven energy injection: the NN was never invoked for IC4 (NN_frac = 0.000). Importantly, the drift is oscillatory throughout—not monotonically increasing—consistent with symplectic shadow-Hamiltonian behaviour, and all bodies remain bounded (IC4 final RMS = 1.60 AU).

Figure 9 shows the energy drift time series for all three ICs.

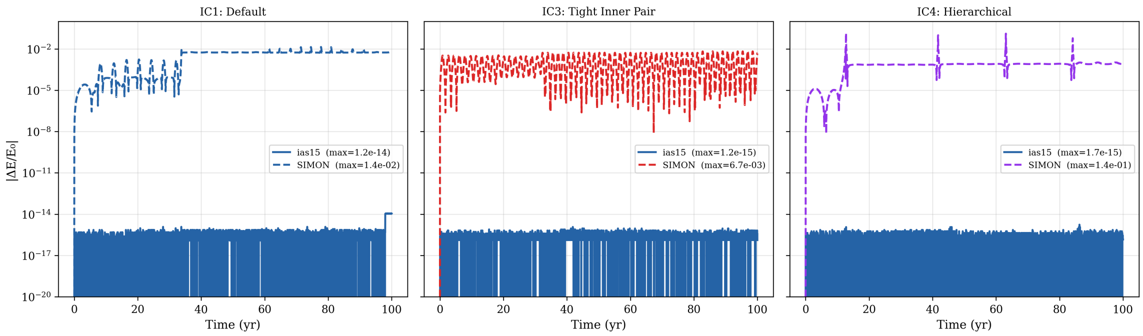

Figure 9: Fractional energy drift $|\Delta E/E_0|$ over 100 years for `ias15` and `SIMON` at $dt = 0.04\,\text{yr}$ across three initial conditions.

5.3.2 Timestep Scaling: Quantitative Confirmation of dt^2 Prediction

Second-order leapfrog theory predicts that energy drift should scale as dt^2: halving the timestep should reduce drift by a factor of four. We test this by running `SIMON` on IC1, IC3, and IC4 at two timestep values ($dt = 0.01\,\text{yr}$ and $dt = 0.04\,\text{yr}$). The fitted log-log slopes are:

- IC1: slope = 1.32 (NN_frac = 0.0045)
- IC3: slope = 2.43 (NN_frac = 0.000)
- IC4: slope = 1.86 (NN_frac = 0.000)
- Average: 1.87 (theoretical prediction: 2.0)

The average slope of 1.87 is consistent with dt^2 scaling, given that only two data points are available per IC. For IC4 specifically, the drift reduces from 13.77% at $dt = 0.04\,\text{yr}$ to 1.05% at $dt = 0.01\,\text{yr}$—a factor of 13.2×, close to the dt^2 prediction of 16×. IC3 and IC4 follow this scaling despite NN_frac = 0.000 at both timesteps, confirming that their energy drift is controlled by the leapfrog timestep rather than by the neural correction.

Figure 10 shows the log-log scaling plot.

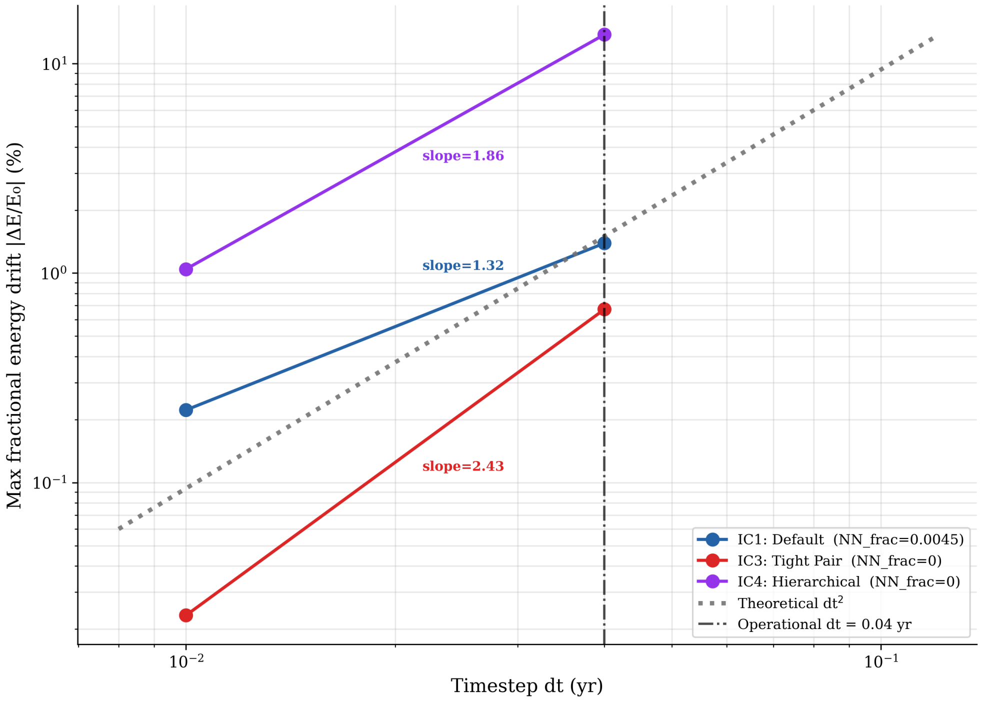

Figure 10: Maximum fractional energy drift vs. timestep for IC1, IC3, and IC4. The grey reference line shows the expected dt^2 leapfrog scaling.

5.3.3 Training Seed Robustness

To verify that results from the seed-42 model used throughout this paper are not specific to a particular random initialisation, training was repeated five times with seeds $\{42, 123, 456, 789, 1000\}$. Table 6 summarises the training and simulation outcomes.

Table 6: Training seed robustness for the full SIMON run on IC1 at $T = 100\,\text{yr}$ and $dt = 0.04\,\text{yr}$.

Seed	**Val. MSE**	$\boldsymbol{\lambda}$ (yr^{-1})	**Bounded**
42 (primary model)	1.11×10^{-5}	0.1655	Yes
123	1.72×10^{-5}	0.1655	Yes
456 (worst)	1.75×10^{-5}	0.1655	Yes
789	1.68×10^{-5}	0.1655	Yes
1000	1.43×10^{-5}	0.1655	Yes
Mean	1.54×10^{-5}	0.1655	5/5
CV	15.8%	0%	—

The divergence rate $\lambda = 0.1655\,\text{yr}^{-1}$ is identical to four decimal places across all five seeds. All five independently trained models produce bounded dynamics with identical fitted divergence rate, confirming that downstream stability is insensitive to random initialisation.

The 15.8% coefficient of variation in validation MSE does not imply dynamical sensitivity to seed choice: the variation is concentrated at sub-softening distances ($r < 5 \times 10^{-4}$ AU) where the safety gate bypasses NN predictions entirely. In the operationally relevant range ($r \geq 10^{-3}$ AU), all seeds produce correction factor errors below 0.3%—functionally identical—and the simulation metrics are invariant.

Figure 11 shows the training convergence and per-seed validation MSE.

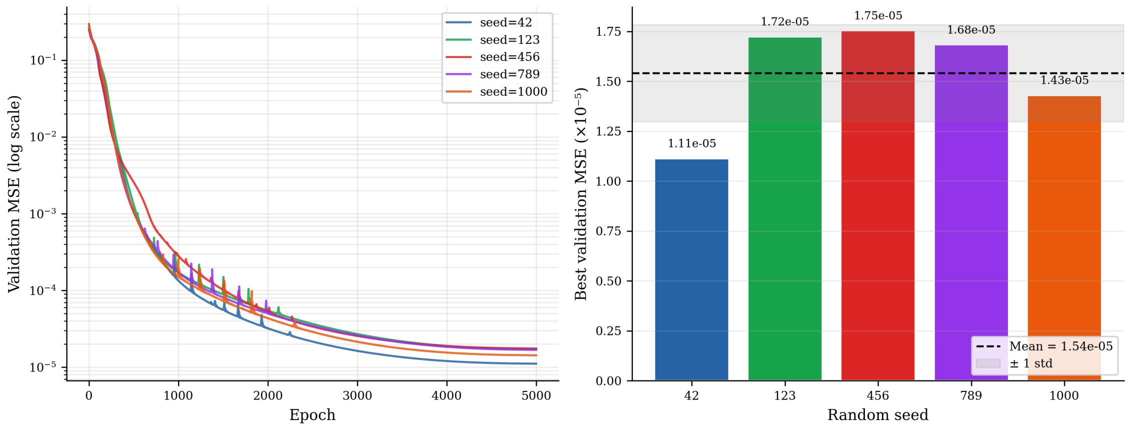

Figure 11: Training seed robustness across five random initialisations. Left: validation MSE curves; right: best validation MSE per seed.

5.4 Results IV — Computational Performance

5.4.1 Speed-Accuracy Frontier

In this measurement, we observe the final position deviation of the bodies predicted by `SIMON`, in comparison with `ias15`, over a 100 year time horizon. The x-axis is the throughput, which is how often the current state of the system is sampled, `Rebound`, and the y-axis is RMS position error, on a logarithmic scale. The total number of samples is fixed across all simulations. The timestep dt is the timestep size the hybrid model uses internally. Points toward the lower-right of the graph are better (higher throughput, lower error). Throughput is computed as $\frac{n_{\text{samples}}}{t_{\text{total}}}$.

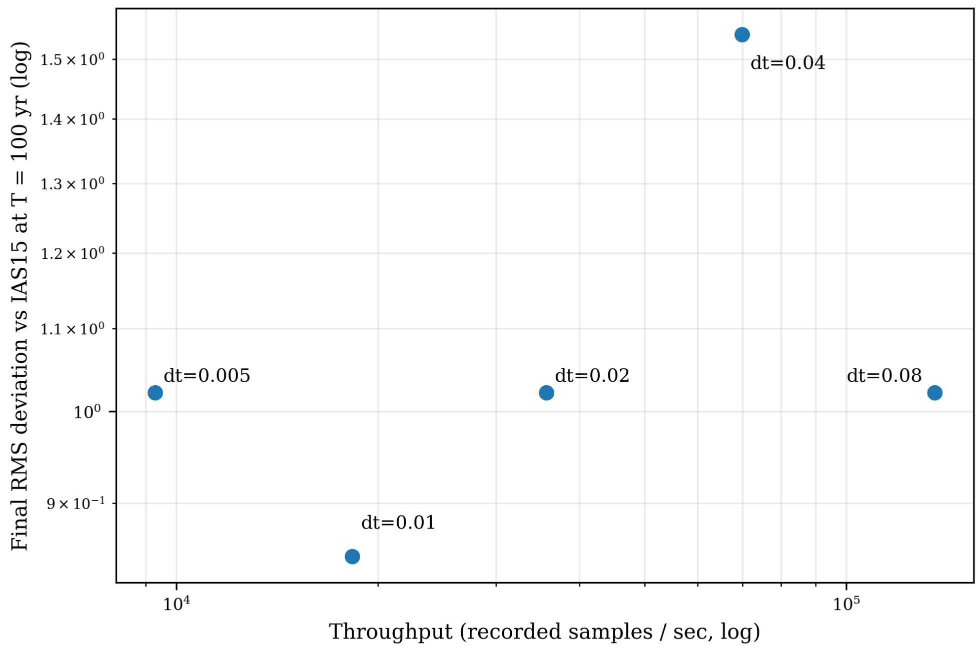

Figure 12: Speed–accuracy frontier for `SIMON` across timestep values. Lower-right points indicate higher throughput and lower final RMS error.

The `ias15` baseline completes in 0.080 seconds. `SIMON` outperforms `ias15` at $dt = 0.04$ (1.12× speedup) and $dt = 0.08$ (2.18× speedup). The lowest final RMS error is 0.847 at $dt = 0.01$. This is the optimal point for accuracy, but it is approximately 3.4× slower than `ias15`. The errors at $dt = 0.005$, 0.02, and 0.08 are all approximately 1.02, reflecting the chaotic saturation scale. After 100 years, any two different integrations of the same system will diverge to roughly one orbital radius, regardless of timestep. $dt = 0.04$ shows a slightly higher error of 1.54, which is a chaotic fluctuation (a different phase of the orbit was reached at $T = 100$).

Table 7 provides the full quantitative results for the frontier sweep.

Table 7: Speed–accuracy frontier results for `SIMON` over $T = 100\,\text{yr}$. The bold row marks the operational timestep.

dt (yr)	Final RMS (AU)	Throughput (samp/s)	Time/step (μs)	Total time (s)	Steps	Speedup
0.005	1.022	9,285	26.9	0.538	20,000	0.15×
0.010	0.847	18,307	27.3	0.273	10,000	0.29×
0.020	1.022	35,672	28.0	0.140	5,000	0.57×
0.040	**1.543**	**69,880**	**28.6**	**0.072**	**2,500**	**1.12×**
0.080	1.022	135,629	29.5	0.037	1,250	2.18×

The per-step time ranges from $26.9\,\mu$s at $dt = 0.005$ to $29.5\,\mu$s at $dt = 0.08$: a range of only $2.6\,\mu$s across a 16× change in timestep. This nearly flat profile indicates that the cost of an individual `SIMON` step is approximately constant and is dominated by force evaluation, pairwise distance checks, and occasional adaptive sub-stepping. Therefore, the observed runtime improvement at larger dt comes primarily from taking fewer macro steps, not from making each individual step substantially cheaper. For a fixed $T = 100\,\mathrm{yr}$ horizon, the number of macro steps scales approximately as T/dt, so total runtime scales roughly as (T/dt) times the near-constant per-step cost.

6 Discussion and Conclusion

6.1 The Two-Constraint Principle

The central finding of this work is that long-horizon orbital stability in a hybrid neural three-body integrator is not produced by the neural network alone. It requires two complementary physical constraints: adaptive sub-stepping and analytic force direction. Adaptive sub-stepping operates at the integration level by resolving close encounters that a fixed macro-timestep can under-resolve. Analytic force direction operates at the force-computation level by enforcing the exact Newtonian radial symmetry at every step.

The ablation study shows that these constraints have distinct roles. Removing adaptive sub-stepping causes an under-resolved close encounter to produce ejection in the timestep frontier. Replacing analytic force direction with a learned direction also leads to unbounded motion for the tested initial condition. By contrast, removing the neural correction while retaining both physical constraints preserves bounded dynamics, with only a modest increase in final RMS error. Thus, the neural network correction is best understood as an accuracy refinement, while adaptive sub-stepping and analytic direction are the main stability mechanisms.

The broader implication is that neural models for chaotic physical systems should not freely learn quantities whose symmetries are already known exactly. In `SIMON`, Newtonian radial direction is enforced analytically, and the network is restricted to a low-dimensional scalar correction during close encounters. This suggests a general design principle for hybrid scientific machine learning: preserve exact physical structure where available, and use learning only for targeted corrections where analytic or numerical approximations are most fragile.

6.2 Positioning `SIMON` Relative to `ias15`

`SIMON` should not be interpreted as a replacement for `ias15`. The reference integrator remains substantially more accurate in energy conservation because it uses high-order adaptive Gauß–Radau integration. `SIMON`, by contrast, uses a second-order leapfrog backbone with targeted sub-stepping, so its energy drift is larger and follows the expected timestep-dependent behaviour of the underlying integrator.

The relevant comparison is therefore not whether `SIMON` reproduces every `ias15` coordinate exactly over 100 years. In a chaotic three-body system, small model or numerical differences naturally accumulate as phase drift. The more meaningful question is whether the hybrid method preserves the qualitative dynamical regime: bounded configurations should remain bounded, physically unstable configurations should be identified as unstable, and energy drift should remain physically interpretable.

Under this interpretation, `SIMON` identifies a stability-preserving operating point. At the operational timestep $dt = 0.04\,\mathrm{yr}$, it preserves bounded dynamics across all tested stable configurations while providing a modest computational-cost reduction relative to `ias15`. Coarser timesteps can give larger speedups, but they are best interpreted as part of the speed–accuracy frontier rather than as the main operating regime.

To answer the research question:

> *How closely can physically constrained neural network force models approximate the long-term stability of traditional numerical integrators in three-body simulations, while reducing computational cost without sacrificing qualitative dynamical stability?*

In the tested three-body regimes, physically constrained neural force correction can approximate the qualitative long-horizon behaviour of traditional numerical integration. All configurations that remain bounded under `ias15` also remain bounded under `SIMON`, and the configurations that eject under `ias15` are also identified as physically unstable by `SIMON`. The main conclusion is therefore not that `SIMON` replaces `ias15`, but that neural force correction can be made dynamically usable in chaotic gravitational simulation when exact physical structure is enforced analytically.

6.3 Limitations and Future Work

This work should be interpreted as a controlled methodological study rather than as a universal neural replacement for high-order reference integrators. Several limitations define the scope of the present results and suggest natural directions for future work.

First, the ablation study in Section 5.1 is centred on IC1. Although the multi-IC evaluation in Section 5.2 provides evidence that the full `SIMON` configuration generalises across several regimes, the causal decomposition of adaptive sub-stepping, analytic direction, and neural correction should be repeated over a broader set of mass ratios and initial geometries.

Second, the fitted divergence-rate λ is a proxy rather than a formal Lyapunov exponent. It compares `SIMON` against `ias15`, so it reflects both intrinsic chaotic divergence and accumulated model mismatch. Future work could compute formal Lyapunov exponents by evolving nearby initial conditions under the same dynamics and comparing them with the model-to-reference divergence used here [Bishop, 2025].

Third, the operating timestep and close-encounter thresholds are tuned for the tested three-body systems. Larger systems, different mass ratios, or more extreme eccentricities may require different thresholds, broader training distributions, or additional safeguards. The pairwise structure of the correction model should extend naturally to $N > 3$ systems, but this has not yet been demonstrated.

Fourth, the present implementation is not optimised for maximum computational speed. The measured speedup is modest because runtime includes force evaluation, gating, sub-stepping, and framework overhead. Future implementations could improve throughput through vectorised pair evaluation, batching, GPU inference, or lower-level compiled kernels.

Taken together, these limitations do not weaken the central result; they define its proper scope. `SIMON` demonstrates that learned force correction can be useful in chaotic gravitational simulation when the physically dangerous parts of the problem—force direction and close-encounter resolution—are constrained analytically rather than left to an unconstrained neural approximation.

References

Peter Battaglia, Razvan Pascanu, Matthew Lai, Danilo Jimenez Rezende, and Koray Kavukcuoglu. Interaction networks for learning about objects, relations and physics. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, 2016.

Robert Bishop. Chaos. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, Fall 2025 edition, 2025.

Sylvio R. Bistafa. Euler's three-body problem. *Euleriana*, 1(2):181–187, 2021. doi: 10.56031/2693-9908.1017. URL https://scholarlycommons.pacific.edu/euleriana/vol1/iss2/6.

Tjarda Boekholt and Simon Portegies Zwart. On the reliability of N-body simulations. *Computational Astrophysics and Cosmology*, 2(1):2, 2015. doi: 10.1186/s40668-014-0005-3. URL https://doi.org/10.1186/s40668-014-0005-3.

Philip G. Breen, Christopher N. Foley, Tjarda Boekholt, and Simon Portegies Zwart. Newton versus the machine: Solving the chaotic three-body problem using deep neural networks. *Monthly Notices of the Royal Astronomical Society*, 494(2):2465–2470, 2020. doi: 10.1093/mnras/staa713. URL https://doi.org/10.1093/mnras/staa713.

Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://arxiv.org/abs/1806.07366.

Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks, 2020. URL https://arxiv.org/abs/2003.04630.

Salvatore Cuomo, Vincenzo Schiano Di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific machine learning through physics–informed neural networks: Where we are and what's next. *Journal of Scientific Computing*, 92(3):88, 2022. doi: 10.1007/s10915-022-01939-z. URL https://doi.org/10.1007/s10915-022-01939-z.

Marco David and Florian Méhats. Symplectic learning for hamiltonian neural networks. *Journal of Computational Physics*, 494:112495, 2023. doi: 10.1016/j.jcp.2023.112495. URL https://doi.org/10.1016/j.jcp.2023.112495.

Walter Dehnen. Towards optimal softening in three-dimensional N-body codes — I. Minimizing the force error. *Monthly Notices of the Royal Astronomical Society*, 324(2):

273–291, 06 2001. doi: 10.1046/j.1365-8711.2001.04237.x. URL https://doi.org/10.1046/j.1365-8711.2001.04237.x.

Jonathan P Gardner, John C Mather, Mark Clampin, Rene Doyon, Matthew A Greenhouse, Heidi B Hammel, John B Hutchings, Peter Jakobsen, Simon J Lilly, Knox S Long, et al. The james webb space telescope. *Space Science Reviews*, 123(4):485–606, 2006.

Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1906.01563.

Nate Gruver, Marc Anton Finzi, Samuel Don Stanton, and Andrew Gordon Wilson. Deconstructing the inductive biases of hamiltonian neural networks. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=EDeVYpT42oS.

Ernst Hairer, Christian Lubich, and Gerhard Wanner. *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. Springer, 2 edition, 2006. doi: 10.1007/3-540-30666-8. URL https://doi.org/10.1007/3-540-30666-8.

Philipp Horn, Veronica Saz Ulibarrena, Barry Koren, and Simon Portegies Zwart. A generalized framework of neural networks for hamiltonian systems. *Journal of Computational Physics*, 521:113536, 2025. doi: 10.1016/j.jcp.2024.113536. URL https://doi.org/10.1016/j.jcp.2024.113536.

Pengzhan Jin, Zhen Zhang, Aiqing Zhu, Yifa Tang, and George Em Karniadakis. SympNets: Intrinsic structure-preserving symplectic networks for identifying hamiltonian systems. *Neural Networks*, 132:166–179, 2020. doi: 10.1016/j.neunet.2020.08.017. URL https://doi.org/10.1016/j.neunet.2020.08.017.

Joseph-Louis Lagrange. Essai sur le problème des trois corps. *Oeuvres de Lagrange*, 3:229–331, 1772.

Alec J. Linot, Joshua W. Burby, Qi Tang, Prasanna Balaprakash, Michael D. Graham, and Romit Maulik. Stabilized neural ordinary differential equations for long-time forecasting of dynamical systems, 2022. URL https://arxiv.org/abs/2203.15706.

Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017. URL https://arxiv.org/abs/1608.03983.

Georgios Margazoglou and Luca Magri. Stability analysis of chaotic systems from data. *Nonlinear Dynamics*, 111:8799–8819, 2023. doi: 10.1007/s11071-023-08285-1. URL https://doi.org/10.1007/s11071-023-08285-1.

Carl D Murray and Stanley F Dermott. *Solar system dynamics*. Cambridge university press, 1999.

Edward Ott. *Chaos in Dynamical Systems*. Cambridge University Press, Cambridge, UK, 2nd edition, 2002. ISBN 978-0521010849.

Manuel Santos Pereira, Luís Tripa, Nélson Lima, Francisco Caldas, and Cláudia Soares. Advancing solutions for the three-body problem through physics–informed neural networks, 2025. URL https://arxiv.org/abs/2503.04585.

Jules Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. divergence des séries de m. lindstedt. *Acta Mathematica*, 13(1):1–270, 1890. doi: 10.1007/BF02392506.

H. Rein and S.-F. Liu. REBOUND: an open-source multi-purpose N-body code for collisional dynamics. *Astronomy & Astrophysics*, 537:A128, January 2012. doi: 10.1051/0004-6361/201118085.

Hanno Rein and David S. Spiegel. ias15: a fast, adaptive, high-order integrator for gravitational dynamics, accurate to machine precision over a billion orbits. *Monthly Notices of the Royal Astronomical Society*, 446(2):1424–1437, November 2014. ISSN 0035-8711. doi: 10.1093/mnras/stu2164. URL http://dx.doi.org/10.1093/mnras/stu2164.

Hanno Rein and Daniel Tamayo. whfast: a fast and unbiased implementation of a symplectic wisdom–holman integrator for long-term gravitational simulations. *Monthly Notices of the Royal Astronomical Society*, 452(1):376–388, July 2015. ISSN 1365-2966. doi: 10.1093/mnras/stv1257. URL http://dx.doi.org/10.1093/mnras/stv1257.

Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter W. Battaglia. Learning to simulate complex physics with graph networks. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 8459–8468. PMLR, 2020. URL https://proceedings.mlr.press/v119/sanchez-gonzalez20a.html.

Yunjin Tong, Shiying Xiong, Xingzhe He, Guanghan Pan, and Bo Zhu. Symplectic neural networks in taylor series form for hamiltonian systems. *Journal of Computational Physics*, 437:110325, 2021. doi: 10.1016/j.jcp.2021.110325. URL https://doi.org/10.1016/j.jcp.2021.110325.

Veronica Saz Ulibarrena, Philipp Horn, Simon Portegies Zwart, Elena Sellentin, Barry Koren, and Maxwell X Cai. A hybrid approach for solving the gravitational n-body problem with artificial neural networks. *Journal of Computational Physics*, 496:112596, 2024.

Yaofeng Desmond Zhong, Biswadip Dey, and Amit Chakraborty. Symplectic ODE-net: Learning hamiltonian dynamics with control. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1909.12077.